

L13b: Interpretable Allocation and Model-Predictive Rebalancing

CHEME 5660 Cornell University

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Objectives for Today

Previously, we chose the share counts of a portfolio once. How do we manage that portfolio every day, and how do we judge it when the real market gives us only one path?

Today's concepts

- Allocation: maximum utility within a budget and trading limits, with a score for each stock from the single index model and the market state.
- Rebalancing: a model-predictive controller that re-plans at the horizon, at a band exit, or at a drawdown stop.
- Evaluation: one historical path, and then typical and stressed synthetic markets from a jump hidden Markov model.

What You Need to Know

Core concepts

- Each rebalance maximizes a utility $U(n;\gamma)$ within a budget and four trading limits. The score γ_i has the sign of the stock's expected growth rate, and the market state ξ changes only its size.
- The book re-plans when 21 trading days have passed, when its wealth leaves the 95% band of a forecast, or when it falls 8% below its peak.
- A backtest is one historical path. A market generator gives many typical futures and sell-offs on which to test the same method.

For understanding (you won't reproduce these)

- The four utilities, the three benchmark weighting rules, and the hedge pool.
- The daily single index update, the forecast covariance, and the trading costs.
- The ten selection rules, and how the typical and stress sets are chosen.

Lectures and Examples

Lecture:

CHEME-5660-L13b-Lecture-MPC-Allocation-Fall-2026.ipynb

Examples:

CHEME-5660-L13b-Example-MPC-RealData-Fall-2026.ipynb

CHEME-5660-L13b-Example-MPC-SyntheticData-Fall-2026.ipynb

Live Portfolio Management

Live portfolio management couples three problems:

- Allocation (what to hold): maximize a utility of a score γ_i for each stock, within trading limits.
- Rebalancing (when to trade): re-plan at a 21-day horizon, at a band exit, or at a drawdown stop.
- Market generator (evaluation): a backtest is one historical path. A jump hidden Markov model (JumpHMM) generates typical and stressed futures.

The study uses the 373 S&P 500 stocks with a complete history. It calibrates on 2014 to 2024 and trades from January 2, 2025 to September 4, 2026, 420 trading days.

Study: J. D. Varner, INFORMS Annual Meeting 2026, arXiv:2611.03022.

Maximize Utility Subject to Trading Limits

A selection rule picks a basket $\mathcal{N}$ of K stocks. Each rebalance chooses the shares n_i of each stock, with weight $w_i = n_i S_i / W$, to solve:

$\underset{n \geq 0}{\text{maximize}}$	$U(n;\gamma)$	(utility)
subject to	$\sum_{i \in \mathcal{N}} n_i S_i \leq W$	(budget)
	$w_i \leq \bar{w}$	(single-name cap)
	$\sqrt{w^\top \Sigma_g \, w} \leq \sigma_{\text{SPY}}$	(volatility cap)
	$\bar{c} \, \lVert n - n^{\text{prev}} \rVert_1 \leq B$	(turnover budget)
	$\beta^\top w \leq \bar{\beta}$	(β cap, hedge on)

where S_i is the close price, W is the wealth, and γ_i is the score of stock i.

The Trading Limits

Limit	What it does	Setting in the study
Single-name cap $\bar{w}$	No stock is more than $\bar{w}$ of the wealth.	$\bar{w} = \max(0.10,\ 1.5/K)$
Volatility cap σ_{SPY}	The book is no more volatile than `SPY`. Σ_g: covariance of daily growth rates.	`SPY` over 2014 to 2024, 17.3% a year
Turnover budget B	A rebalance spends at most B on the spread. $\bar{c}$: half spread per share.	$B = 0.10\,W$, half spread of 5 basis points
β cap $\bar{\beta}$	The book's β to `SPY` is at most $\bar{\beta}$.	$\bar{\beta} = \max(0,\ 1 - \xi)$

The β cap applies only when the market hedge is on. It tightens as the market state ξ turns bearish. The book can then also hold up to five stocks whose β is zero or negative, the hedge pool.

The Weighting Rules

Four utilities use the stocks with a positive score, $\gamma_i > 0$:

Utility	$U(n;\gamma)$	Solution with the budget alone
Cobb–Douglas	$\sum_i \gamma_i \ln n_i$	$w_i \propto \gamma_i$
CES on shares	$\left(\sum_i \gamma_i \, n_i^{\rho}\right)^{1/\rho}$	$n_i \propto (\gamma_i / S_i)^{\eta}$
CES on weights	$\left(\sum_i \gamma_i \, w_i^{\rho}\right)^{1/\rho}$	$w_i \propto \gamma_i^{\eta}$
Stone–Geary	$\sum_i \gamma_i \ln (n_i - b_i)$	$(n_i - b_i)\, S_i \propto \gamma_i$

where $\rho = 1 - 1/\eta$ with $\eta = 0.5$, and $b_i S_i = 0.01\, W$. With the trading limits there is no closed form. Each problem is a convex cone program.

Three benchmarks run under the same limits: minimum variance, the tangency portfolio, and a 130/30 long–short book. That makes seven weighting rules.

The Score γ

Each stock gets a score after every close:

$$\gamma_i = \tanh\left(\frac{\alpha_i + \beta_i\,\tilde{g}_M}{\max\left(|\beta_i|,\,0.1\right)^{\xi}}\right)$$

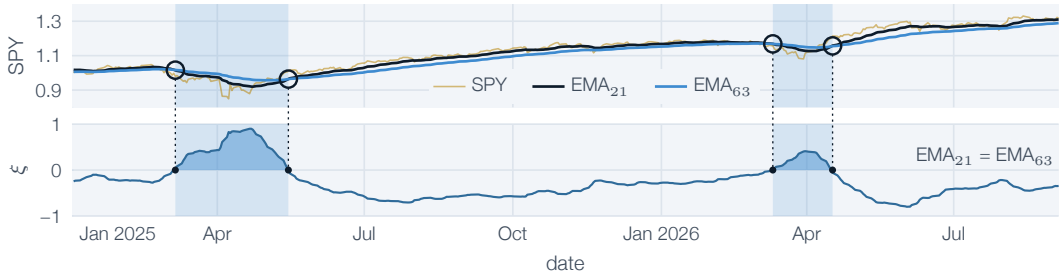
- Expected growth rate (numerator): (α_i, β_i) are the single index model parameters of stock i against `SPY`, updated daily, and $\tilde{g}_M$ is the 252-day EMA of the daily growth rate of `SPY` (1/yr).
- Penalty on β (denominator): set by the market state ξ, which compares the 21-day and 63-day EMA of the `SPY` close:

$$\xi = -G\left(\frac{\text{EMA}_{21}}{\text{EMA}_{63}} - 1\right)$$

where $G = 21.23$ makes $|\xi| \leq 1$ on 99% of the days from 2014 to 2024.

The Market State

The market state is positive when $\text{EMA}_{21} < \text{EMA}_{63}$, as in a falling market:



The shaded days have $\xi > 0$: 75 of the 420 trading days, in spring 2025 and in March and April 2026. Each ring marks a day on which the two EMAs cross, where $\xi = 0$.

Reading the Score

- Sign: the penalty is positive, so the score has the sign of the expected growth rate.
- Size: in a falling market ($\xi > 0$), the score favors stocks with a low β. In a rising market ($\xi < 0$), it favors stocks with a high β.
- Bounds: the hyperbolic tangent keeps the score between -1 and 1, and the floor of 0.1 keeps the penalty finite when β_i is near zero.

This score is the preference weight we used previously, in a form that works for every stock. It does two jobs:

- Weighting: stocks with $\gamma_i > 0$ are held long, and the 130/30 rule shorts stocks with $\gamma_i < 0$.
- Selection: a rule picks the stocks with the highest scores.

Rebalancing: The Forecast Band

A rebalance sets target shares, and the plan is to hold them. We call each rebalance a re-plan. The method uses model-predictive control:

- Forecast: a multi-asset geometric Brownian motion, 1,000 paths, 21 trading days ahead, with mean growth rates $\alpha_i + \beta_i \, \tilde{g}_M$ and covariance Σ_g.
- Plant: the market. The measurement is the wealth W of the book at the real close prices.

With $\mu_W(\tau)$ and $\sigma_W(\tau)$ the mean and standard deviation of the forecast wealth τ trading days after the re-plan, the 95% forecast band is:

$$\mu_W(\tau) - 1.96\,\sigma_W(\tau) \;\le\; W \;\le\; \mu_W(\tau) + 1.96\,\sigma_W(\tau)$$

The Rebalancing Triggers

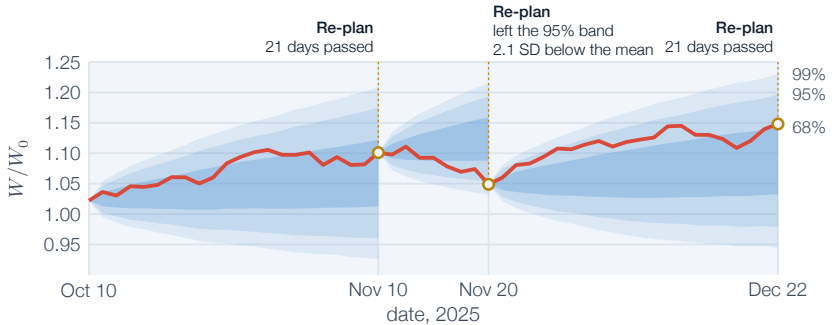
After each close, the controller re-plans on the first trigger that holds:

1. Drawdown stop: the wealth is more than 8% below its running peak. The peak then resets, so the stop fires once for each fresh 8% loss.
2. Horizon: the plan has run for 21 trading days.
3. Band exit: the wealth has left the 95% band of the forecast.

A re-plan solves the same allocation problem from the shares held, with that day's estimates. No trigger changes the utility or the limits. Between re-plans, the book does not trade.

One Episode of the Real-Data Backtest

The red line is the wealth of one book over its starting wealth W_0. Each gold ring is a re-plan, and each re-plan starts a new forecast:



Over the whole window, this book re-planned 27 times: 17 times at the horizon, 7 times at a band exit, and 3 times at a drawdown stop.

Fixed Schedule and Trading Costs

Do the triggers beat a fixed schedule?

We reran everything on a fixed 21-day schedule. The ending wealth is the same. The triggers take about a point off the worst drawdown, on real prices and in simulated sell-offs.

What does a trade cost?

Orders fill at the next open, sells first and then buys. Each trade pays a half spread of 5 basis points, plus an impact cost that grows with the fraction of the stock's mean daily volume that the order takes. There is no commission.

The Selection Rules

Each run pairs a selection rule with a weighting rule, with the hedge on or off. The study compares ten selection rules in three groups:

- 33 stocks, 3 per sector: in each of the 11 sectors, the 3 highest scores, the top 3 stocks of a bandit, or 3 stocks at random.
- 13 stocks: the 13 highest scores in the universe, with or without a correlation screen or a liquidity screen, or 13 stocks at random.
- Hand-picked: 13 stocks chosen in advance, without 2025 data, and never re-picked. A second list adds two more stocks.

Every other rule picks on the first day, every 21 trading days after that, and at every band exit.

Selection Matters More Than Weighting

Real-data backtest. Each row is a selection rule, and each dot is a weighting rule:



With the hedge off, seven weighting rules on one basket ended between 1.09 and 1.23 times the initial wealth. Different baskets ended between 1.03 and 1.70.

Example: The Method on Real Data

Example:
CHEME-5660-L13b-Example-MPC-RealData-Fall-2026.ipynb

In this example, we calibrate the method on 2014 to 2024 for the 373 S&P 500 stocks with a complete history and trade 2025 and 2026. We follow the market state and the scores, run one book day by day and read why it re-planned, and then change the weighting rule, the selection rule, the hedge, and the basket on the same path.

Market Generator: JumpHMM

On real data, the method runs once, on one path with one broad sell-off in the spring of 2025. Previously, we built a market generator from jump hidden Markov models (JumpHMM). The daily growth rate of synthetic stock i in sector s is:

$$g_i(t) = \underbrace{\alpha_i}_{\text{trend}} + \underbrace{\beta_i(t)\, g_m(t)}_{\text{market: 1 JumpHMM}} + \underbrace{\delta_i\, f_s(t)}_{\text{sector: 11 JumpHMMs}} + \underbrace{\psi_i\, \tilde{\varepsilon}_i(t)}_{\text{own: 373 JumpHMMs}}$$

- The generator is fitted to the same 373 stocks on 2014 to 2024, and each $\beta_i(t)$ drifts around the stock's estimate.
- It reproduces the fat tails of real growth rates, but its volatility clustering fades after a day.
- The controller still forecasts with a geometric Brownian motion. The test markets have fat tails that the forecast model does not have.

Typical and Stress Sets

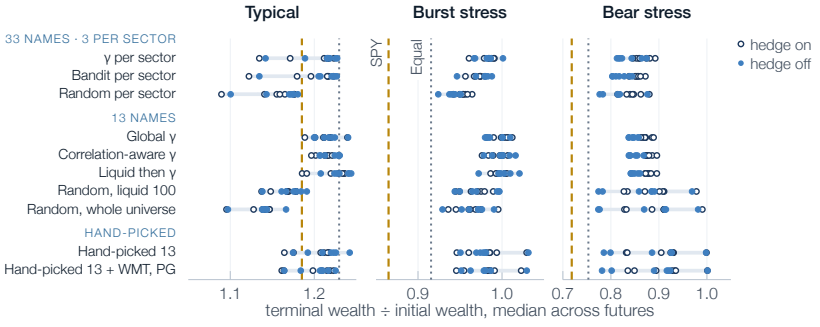
On each synthetic market, the method calibrates on 2014 to 2024 and trades January 2025 to September 2026 with the real settings. We draw 10,000 paths of the market model and rank them by the maximum drawdown of synthetic `SPY`:

- Typical: 600 paths drawn at random. The median drawdown of `SPY` is 17.6%.
- Burst stress: the 200 deepest drawdowns among the paths with a jump burst in the trading window. The median drawdown is 36.4%.
- Bear stress: the 200 deepest drawdowns among the paths without a jump burst. These declines are slower and more sustained, with a median of 40.6%.

Some paths are in more than one set, so the three sets hold 978 markets.

Every Median Ends Above SPY

Synthetic futures. Each dot is the median across the futures of a set:



In both stress sets, every median ended above SPY and above the equal-weight portfolio. The random selection rules did as well.

Reading the Synthetic Results

- The margin: it came from parts of the method that every book shares, not from stock selection or from the timing of the re-plans.
- Selection by score: in typical futures with the hedge off, 13 stocks picked by score beat 13 random stocks in 62 to 71% of the futures. In bear futures, random picks did as well or better for most weighting rules.
- The generator's trends: the generator keeps each stock's 2014 to 2024 trend, which the score estimates. Part of the advantage of the score may not carry over to real prices.
- The hedge: it reduced terminal wealth on real prices. It added to terminal wealth in the bear stress set, whose declines were the most sustained.

Example: The Method on Synthetic Markets

Example:
CHEME-5660-L13b-Example-MPC-SyntheticData-Fall-2026.ipynb

In this example, we draw synthetic markets from a generator fitted to the same 373 stocks, and choose typical markets and two kinds of sell-off as the study does. We then run the method on each market, and compare the books with `SPY` and an equal-weight portfolio set by set.

Summary

We followed one system for managing a stock portfolio every day, from the allocation problem to its test on real and synthetic markets.

Key takeaways

- Allocation: stock selection mattered more than the weighting rule. Seven weighting rules on one basket ended close together, while different baskets ended far apart.
- Rebalancing: re-plans came at the horizon, a band exit, or a drawdown stop. The triggers took about a point off the worst drawdown.
- Evaluation: in both sell-off sets, every median ended above `SPY`. The generator keeps each stock's past trend, which favors selection by score.

A paper-trading version of this system is running now.